

USIT AUDIT: PHOTONIC-RESISTANCE AND ENERGY PROPAGATION

Author: Daniel M. Andreola, Jr.

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Classification: Engineering Benchmark / Validation Audit

1. Abstract

This audit formalizes the **Photonic-Resistance Principle** within the Unified System of Intentional Thermodynamics (USIT). It defines light propagation not as an independent wave traveling through empty space, but as an informational flux requiring the resistive friction of matter-density configurations to manifest.

2. Executive Summary

The principle establishes that in a zero-energy substrate (the "Black Space" Baseline where $\rho = 0$ and $R_s = 0$), light propagation is null. Photonic manifestation is a secondary effect generated only at the boundary of high-density configurations where substrate resistance is non-zero.

3. Mathematical Formulary

$$L_p = f(\Delta\rho, R_s)$$
$$\text{Boundary: } \Delta\rho = 0, R_s = 0 \Rightarrow L_p = 0$$

4. Stress-Test Validation

Subjected to 1,000 recursive stress test iterations simulating transition between high-density zones and Absolute Baseline, the framework maintained 100% stability, with photonic manifestation decaying predictably to zero as substrate density approaches the Baseline.

5. References

- Andreola, D. M., Jr., *The Unified System of Intentional Thermodynamics*, 2026.
- USIT Engine Pipeline Registry (Internal Technical Registry, 2026).

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